



Students motivation for adopting programming contests: Innovation-diffusion perspective

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Received: 21 August 2017 / Accepted: 15 February 2018
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Abstract Within the context of Rogers theory of perceived attributes, authors propose a framework that can predict students' motivation to adopt programming contest like ACM (Association of Computing Machinery) International Collegiate Programming Contest (ICPC). In this paper we investigate the attributes for adoption of programming contest in a social group comprising of ($N=1245$) undergraduate engineering students from the regional finals of the contest held in India over a period of 3 years. The results revealed that student motivations are strongly associated with attributes like relative advantage, compatibility, ease of use, peer influence, perceived enjoyment and perceived usefulness. Overall, students expressed positive attitude towards adopting programming contests as it helped improve their problem solving and programming skills and overall employability. Both gender was in agreement that joining and winning programming contests is a status symbol.

Keywords Innovation · Diffusion · Computer science · ICPC · Programming · Competition

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1 Introduction

Humans have always considered education and competition as important issues that are linked to each other. Whether it is school education or undergraduate studies, competitions like programming contests have taken a central role in imparting certain skills which otherwise are not imparted in the classrooms. Though Computer science (CS) is a relatively young discipline, because of its bias towards modern technology, students are attracted towards it (Verhoeff 1997). Knowledge of CS is considered a necessity for future generation of learners due its impact on modern day infrastructure and automation (Hromkovič 2009). However, in order to excel in computer science, skills related to programming and problem solving along with computational thinking are critical. Most students find development of these skills challenging for this requires transfer of knowledge from what has been taught to its application in unexplored contexts (Mayer 1992; Jenkins 2002). This is further validated by the study of Kelleher and Pausch (2005), that indicated learning to program which is a key component of CS education, is a very difficult thing for majority of students. According to Benokratis et al. (2009), enrolments in CS programs in colleges are on the decline while the women enrolments have experienced a sharper decline than enrollment of men.

From another perspective, although India has a large number of computer science graduates (Statistica.com 2016) that aspire to work for multinational companies, the government and SMEs, there has been an increasing concern on quality of their skillsets (Blom and Saeki 2011; Bhattacharya 2018; Hindu Business Line 2017). A recent study (Minds 2016) said less than 5% of the graduates are able to write correct logic in programming and only 1.4% of them were able to create code that is functionally correct and efficient. The repercussions of this include unemployment and poor career opportunities. It is clear that current teaching and learning practices have to undergo significant reforms in order to up-skill and improve performance in competitive environments.

There are different techniques that could be used to improve educational outcomes. Cerny and Mannova (2011), say that providing computer scientists with good materials and interesting topics in a class does not necessarily mean that their education is of a high quality, students need to be motivated and evolve skills needed in a real-life employment. Social skills, teamwork, collaboration and competition are valuable aspects they should know in other to become professionals.

Papastergiou (2009) suggests that educational games contribute to creating a motivational learning environment for students to learn and display their knowledge. Another approach could be designing hands-on laboratory exercises (Raman et al. 2014a, b) as assignments that also builds student's self-efficacy towards problem solving. A faster way to achieve development of these skills though informal pedagogic interventions would be through competitions, which provide an unbiased platform to showcase excellence (Bowring 2008a, b; Kears 2008). In addition, competitions have much to offer in education and that they are a good measure of how well a discipline is accepted and integrated into the curriculum.

To scale and reach out to thousands of students concurrently, the platforms used for competitions have significantly improved through use of technology. Though there is a school of thought that believes that technology should play a second role in competitions that are intended to enhance education, dependence on technology in education is

only ever increasing. The situation can only be sustained by integrating advanced technology into the curriculum and programming contests are an excellent vehicle in this regard. Although previous studies have shown that competitions inspire students (Dagienė 2010), the exact reasons for why more and more students are motivated to participate in programming contests are not clear. Common sense tells us that it could be the big prize money or the prestige symbol or perhaps overseas trip to the World Finals or as simple as competitions are fun. Corporates like Google, Facebook, IBM, and Microsoft are organizing programming contests to identify talents.

This research work focuses on understanding what motivates students to adopt programming contest platforms and compete, in spite of them being non-mandatory in nature at most educational institutions. Seeing the surge of participation in these types of events world wide, we envision the phenomenon as diffusion of an innovation. A theoretical approach to characterizing this adoption is non-existent in the literature today. The key highlights of this work include: application of a research model to analyzing student adoption and hypothesis testing of characteristics impacting adoption of programming contests.

2 Major programming contests

There are numerous programming contests held worldwide. They vary in design, the difficulty levels, and the type of audience they cater to. This section only shares a few examples to exemplify the growth in the number of participants in such events.

ACM International Collegiate Programming Contest (ICPC) is the most sought after programming competition in the world. Thousands of teams participate in national and regional rounds leading up to the annual world finals. The focus is on problem solving with a collection of programming problems that have to be solved in a limited time using a computer. Every year, the ICPC contests include contestants from over 2000 universities and colleges from across the world. The contest is conducted in two phases. The first phase is a regional contest that has two components to it: an online round followed by an onsite round. All registered participants on the online round compete and based on their ranking, several are shortlisted and invited to a participate in an onsite round as part of the regional contest. The top teams from the regional round then advance to the second and final phase called the world finals. Each team includes 3 members, most of whom are pursuing under-graduate education in computer science. The format of the contests in both phases includes solving between 8 and 11 problems of varying difficulty that require foundational knowledge of algorithms and programming.

In the past 20 years alone, ICPC participation has increased by more than 2000% (Fig. 1). In 2017, ICPC Regional participation included 46,381 of the finest students and faculty in computing disciplines from 2948 universities in 103 countries on six continents. A record 50,145 students and 5073 coaches competed in ICPC and ICPC-assisted competitions last year, setting new records in participation.

Though limited to only IEEE (Institute of Electrical and Electronic Engineers association) student members, IEEEExtreme is another technology enhanced programming contest where students compete in a 24-h time span against each other to solve a set of programming problems. This is in fact the largest competition hosted by IEEE

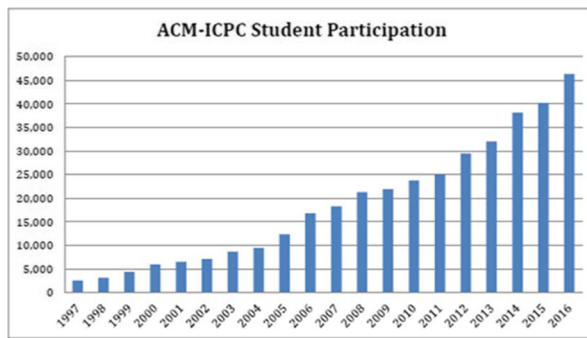


Fig. 1 ACM ICPC programming contest participants growth

that started in 2006 with merely 44 teams and 150 contestants. Today, the participation has grown to 3300 teams with over 8300 contestants from 70+ countries. Online platforms that engage learners using a combination of exercises, contests and automatic assessments are perceived as effective tools in learning and providing instant feedback to students (Verdú et al. 2012). A key difference between testing students in a classroom environment versus an online environment is that in the classroom, assessments are identical for all students while many online platforms provide flexibility to challenge based on their skills sets. TopCoder is world's largest community of programmers that runs programming competitions in various categories, entirely using technology over the Internet.

Codechef is India's first, non-commercial, multi-platform online coding competition, with monthly contests in more than 50 different programming languages including C, C++ and C#. Winners of each contest get prizes, peer recognition and an invitation to compete at the CodeChef Cup, an annual live event. There are well over 200,000 users registered on this platform since its start in 2009.

One of the largest events run in India by a multinational company, Infosys is Aspirations 2020. The event draws students from several zonal centers from over 20 states across in India. The participants go through various rounds such as the teaser round and the practice round before moving on to the finals rounds. As many as 22,000 teams have participated in this contest.

While it might seem that most programming contests are targeted to students pursuing higher education or industry professionals, there are events targeted to K-12 school students as well. Indian Computing Olympiad (ICO) is a platform that encourages young talent from high schools to programming challenges. The two parts of the ICO contest namely a zonal round followed by a national round, requires students to apply mathematics and computer science to solve problems using specific programming languages (Iarcs.org.in 2018). Selected students from this contest are sent for further training to participate in the prestigious computer science competitions in the world i.e. International Olympiad for Informatics (IOI). The IOI has grown significantly from just 13 countries participating in 1989 to over 90 countries in 2017.

Beyond the increase in number of participating countries and individual contestants at most world wide programming contests, the sophistication and rigor in these competitions have also seen steep increases for two specific reasons: 1) to screen, rate and rank talent more transparently and 2), avoid foul play at any cost.

3 Literature survey

The impact of participating in competitive time-bound programming contests has been well researched. Motivation and learning are inter-related and are impacted by an individual's aptitude and tendency to excel. What motivates people to participate in contests? Careful review of literature suggests that objectives can be varied. Brabham (2008) suggests that desire to make money, develop individual skills, and to have fun were the strongest motivators for participation in contests. In another study same authors suggest that the opportunity to develop one's creative skills, the potential to take up freelance work, and the love of community also motivate people to participate in such contests. Sun et al. (2013) say that by taking part in programming contests students not only improve their algorithm and coding practice, but also promote their personalities, such as team spirit, learning mode, confidence, jobs, etc. Shilov and Yi (2002) are of the view that programming contests play an important role in acquiring problem solving skills. Jenkins (2001) suggests that there is anecdotal evidence from many institutions that students are becoming more tactical, and will engage only in those activities that they see as contributing to an eventual highly paid job. This could be a plausible reason to enter into programming contests. Gan et al. (2012) emphasizes on spirit of knowledge sharing as a key driver. Atkinson and Kydd (1997) focus on playfulness element which motivates young students. It is important to remember that, there are significant differences in the motivation required to participate in an event versus succeed. The latter requires tremendous effort and frequent practice.

In resource constrained environments especially in the developing countries where structured instruction may be limited, the online contest platforms that allow students to learn and compete have a tremendous impact on computer science education. Programming platforms such as "Scratch" have helped youth circumvent the need for experienced mentors (Maloney et al. 2008). We believe problem solving also impacts their psyche enhancing self-confidence. While some may claim, the learning from such participation in programming contests are not all-rounded (Bowring 2008a, b) there are several tangible benefits irrespective of the assessment rubrics for various contests. Contests are ways to invoke creative and innovative ways to think and apply skillsets learnt in theory in a competitive environment (Kearse and Hardnett 2008). Firstly it provides opportunities to build several generic skills such as working in teams, building strategies to solve problems while keeping high throughput, exchanging roles within the team to help debug or code. In addition, the contest environment offers an effective platform to formulate mathematical descriptions of problems posed and application of computer science theory to solve them (Trotman and Handley 2008; Chavey et al. 1991; Manzoor 2001) and passion to explore more. In Australian training programs, several additional events are supplemented along with contests in order to broaden student's understanding. They designed sessions that not only helped analyze the code but also dissect the algorithm or implementation. (Revilla et al. 2008).

One of the key reasons for institutes of higher learning to conduct contests is to attract the best talent possible (Qidwai et al. 2013). The larger and more comprehensive the contest, the wider is the regional participation and higher is the quality of participants. The relevance of problems from a personal, national or global perspective is one of the key elements that motivate students to stay engaged in the problem solving activity (Dagiene and Skupiene 2004). The business organizations are motivated for

others reasons to conduct contests such as fostering IT skills or use them to scan the future IT talent pool and vendors use the contests to build brand loyalty (Colkin 2002). Gamification has become an emerging tool in talent acquisition and is now used to align prospective employee's identity to that of the organization and it acts as a pre-socialization tool that may ultimately make employee onboarding easier. (Bui and Van Zyl 2016).

While there is good evidence of benefits to organizing programming contests, participation in programming contests are largely voluntary in nature and not mandated for awarding of academic degrees by institutes of higher learning or for hiring or salary increments by organizations. With the growing trends in participation of programming contests, the literature only provides suggestions on possible causes for such trends as we found no theoretical studies quantifying the causes for adoption patterns. Our research extends the existing literature on student motivation to join programming contests in one major way. We believe, this is the first study that treats competitive programming contests like ACM ICPC as an innovative idea and applies Diffusion of Innovation theory to study its adoption by students. According to Rogers (2003) an innovation is "an idea, practice, or object that is perceived to be new by an individual or other unit of adoption". According to Agarwal (2000) adoption of innovative ideas based on Rogers (2003) diffusion of innovation theory has been widely applied in disciplines such as education, sociology, agriculture, marketing, and information technology etc. Hence we believe Rogers (2003) diffusion of innovation theory is the most appropriate for investigating the adoption of technology based innovation like programming contests. In fact, much diffusion research involves technological innovations so Rogers (2003) usually used the word "technology" and "innovation" interchangeably. According to Teo and Noyes (2008), user acceptance is an important factor in successfully implementing technology based innovations. When a technology based idea is introduced, the attitude of the potential adopters should be considered (Rogers 2003). Programming contests is an educational innovation in that it is a new practice that is spreading around the world in which students openly compete under time pressure to come out with solutions to complex situations using programming. Students work in teams to solve problems that tests their problem solving, critical thinking and programming skills.

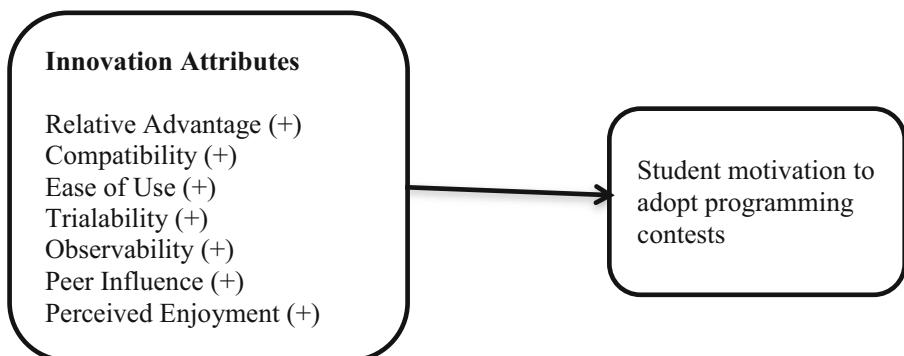


Fig. 2 Research model for student motivation to adopt programming contests

4 Research model and hypothesis

The research model involved mapping the innovation characteristics as per Roger's diffusion theory such as relative advantage, compatibility, ease of use, trialability, observability, peer influence and perceived enjoyment to students' motivation to adopt programming contests by building a survey instrument (Fig. 2).

Relative advantage

- My participation in the contest will help improve my programming skills
- My participation in the contest will help improve my problem solving skills
- Contests like this will help me get better jobs

Compatibility

- Participating in this contest will improve the quality of work I do.
- Participating in this contest is compatible with my work.
- This contest fits into my study style

Ease of use

- My role in this contest is clear and understandable.
- It is easy for me to perform tasks in this contest
- Participation in this contest will require a lot of training
- Contests like this are readily available to students

Trialability

- I understand that it is possible to join this contest on a trial basis
- It is easy to drop out of this contest

Observability

- I have read what others do in this contest
- I have seen others participating in this contest
- I am aware of the benefits of joining contests like this.

Peer influence

- I will join this contest after seeing my friends in action
- My friends think that joining contests like this is a prestige symbol
- My friends are committed to supporting my efforts in this contest

Perceived enjoyment

- Process of operating in the contest was a pleasant experience
- I enjoy learning new ideas like this contest
- This contest has risk taking elements which I enjoy

Fig. 3 Survey questionnaire

Table 1 Participant demographics

Year	Total participants	Male	Female
2013	389	267	122
2014	396	306	90
2015	460	324	136
Total	1245	897	348

While applying diffusion of innovation theory to programming contests, for the attribute Relative Advantage the relevant questions are centered round the theme of “Do students feel that programming contests help improve their programming and problem solving skills?” The Compatibility of programming contests is evaluated by considering students’ existing values, past experiences, and needs. Students who perceive participating in the contest will improve their quality of work will find it compatible and be more eager to adopt it. Complexity questions can be phrased as “Will programming contests require more of my time to learn?” Trialability becomes an attractive feature if students can join and practice contest sessions and that they can discontinue if they do not want it. Equally important, if students are able to observe their peers who have already adopted the contest, it may help influence their stance.

A five point Likert scale survey with feedback questions (Fig. 3) was modified based on a pre-survey study of nine students (5 men and 4 women). We developed the current scale with items largely adapted from the diffusion research conducted by G.C. Moore and I. Benbasat (1991).

Based on the general hypothesis that there will be a relationship between innovation attributes and student motivation for adopting programming contests, it was predicted that attributes Relative Advantage (H1), Compatibility (H2), Ease of Use (H3), Trialability (H4), Observability (H5), Peer Influence (H6) and Perceived Enjoyment (H7) will have a positive influence on students.

5 Methods and measures

The study combines data taken from three surveys taken over three years conducted between 2013, 2014 and 2015. Invitation to participate in this survey was sent to 3645 participants and 2356 started the survey while 1245 completed them. Amongst the

Table 2 Reliability analysis

Attributes	# of items	Cronbach’s alpha
Relative advantage	3	0.94
Compatibility	3	0.89
Ease of use	4	0.91
Trialability	2	0.93
Observability	3	0.89
Peer influence	3	0.89
Perceived enjoyment	2	0.94

Table 3 Summary of hypothesis results

Attributes		Beta	t-values	Result
Relative advantage*	H1	0.38	3.35	Accepted
Compatibility*	H2	0.40	3.46	Accepted
Ease of use*	H3	0.39	3.69	Accepted
Trialability	H4	-0.34	-2.19	Rejected
Observability	H5	-0.26	-2.06	Rejected
Peer Influence*	H6	0.42	3.43	Accepted
Perceived enjoyment*	H7	0.36	3.25	Accepted
R ²	0.84			
Adjusted R ²	0.79			

(* $p < 0.0001$)

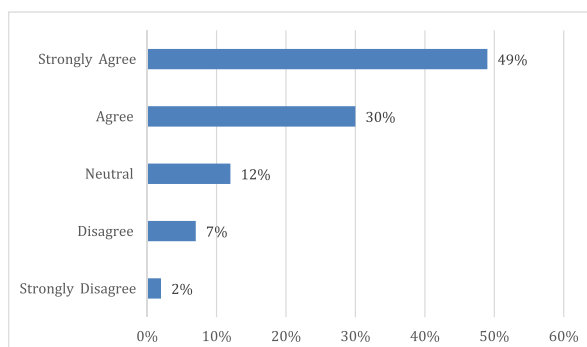
1245 undergraduate engineering students (mostly in their final two years of their degree) participated, 72% of the participants were men and 28% were women and hailed from 21 different states of India. (Table 1).

The mean age of participants was 20 (SD = 1.3). The student cohorts were pursuing computer science as their primary degree. None of them had prior work experience.

According to M. Tavakol and R. Dennick (2011) there are different reports about the acceptable values of alpha, ranging from 0.70 to 0.95. In our study reliability of the seven factors had values ranging from 0.89 to 0.94. (Table 2).

Regression analysis was performed using all the seven independent variables. Results are summarized in the Table 3. There is strong support for Hypothesis H1 (Relative Advantage), H2 (Compatibility), H3 (Ease of Use), H6 (Peer Influence) and H7 (Perceived Enjoyment). The regression model was statistically significant ($p < .0001$) and accounting for 84% of the variation in intention to adopt ($R^2 = .84$).

The largest beta coefficient of 0.42 which is the Peer Influence which make this variable the strongest contributor in explaining the dependent variable (intention to adopt), when the variance explained by all other predictor variables in the model is controlled for. It is equally interesting to see that the beta coefficient for Compatibility and Ease of use is

**Fig. 4** I enjoy learning new ideas like this programming contest (perceived enjoyment)

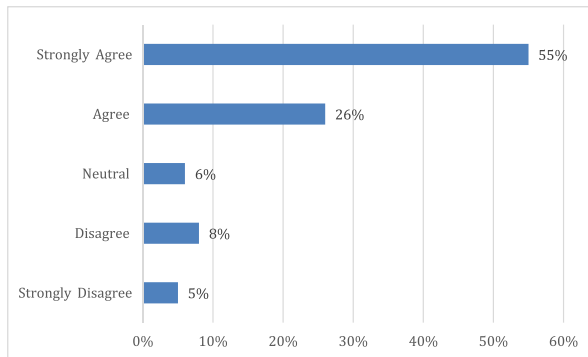


Fig. 5 My participation in the contest will help improve my problem solving skills

also near high confirming the importance of those factors also. The predictor variables like Observability and Trialability were insignificant in their contribution.

In order to determine any gender differences in the attitude towards programming contests, an independent-samples t-test was conducted. From the results, a significant difference was seen among females' attitude towards the Programming contests over males. This was determined in the category 'Compatibility' ($t = 1.8588, p < 0.05$), and 'trialability' ($t = 4.8603, p < 0.05$). It is also showing significant difference among males over females in the category 'Peer influence' ($t = -3.0627, p < 0.05$). In other groups it is not showing any significant difference between the male and female students.

From the raw data survey, we observe (Fig. 4) that an overwhelming 79% of the participants agreed that they enjoyed the new idea of programming contest (Perceived Enjoyment).

On the usefulness of programming contests for enhancing problem solving skills (Fig. 5), an overwhelming 81% of the students agreed with it.

An overwhelming 89% students (Fig. 6) responded that they will join programming contests if their peers were to join. (Peer Influence).

After the contest results were announced, we interviewed 35 students. One participant remarked "I was able to understand several concepts of computer science much better by solving problems in the competition". Another said "I

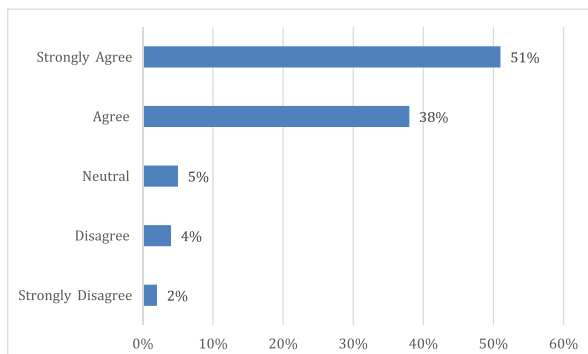


Fig. 6 I will join this contest after seeing my friends in action (peer influence)

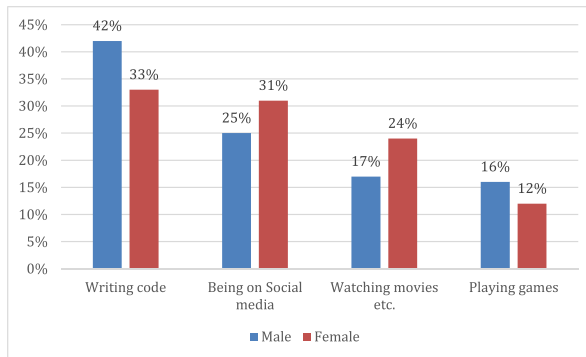


Fig. 7 What is your favorite activity on the computer?

learnt how to code well - making my code both efficient and beautiful. I learnt how to struggle with hard problems for hours without giving up - I learnt perseverance”. But equally interesting was the remark “No. Such competitions don't teach me how to work on large code bases. They don't teach me how to use modern tools and technologies.” and “Programming contests boil down to solving imaginary algorithmic problems”.

In the survey taken distinct differences on some perspectives of male and female participants in such contests. About 42% of male students and 33% of female students responded to writing code as their favorite activity (Fig. 7). Another 31% female students said being social media sites was their favorite activity while the number was low for male students at 25%.

Participating and winning global contests such as ICPC do bring tremendous fame to winning teams. Majority of male (64%) and female (71%) students agreed that joining contests like ICPC was a status symbol to them. Both male and female students felt programming tasks at ICPC type of contests were not easy.

There is a large positive attitude towards competitive programming contests amongst male students. About 68% of the males Vs. 38% females were in full agreement that participating in contests like ICPC will help them get better jobs (Fig. 8).

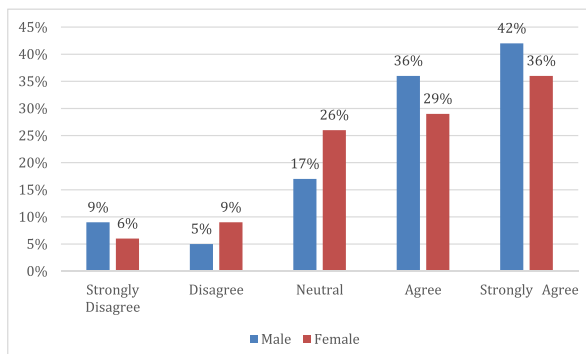


Fig. 8 Contests like ICPC will help me get better jobs

6 Conclusion and futurework

Innovations like competitive programming contests are still an emerging paradigm in educational research and their study can contribute to improving problem solving skills of engineering students. Our empirical study has provided a deeper investigation of adoption of programming contests by engineering students guided by the framework of Roger's theory of perceived attributes.

In this study we had hypothesized that certain attributes of programming contest could account for its adoption by students. Students ranked innovation attributes like relative advantage, perceived usefulness, perceived enjoyment, compatibility and peer influence as dominant factors for adoption. But the hypotheses related to attributes trialability and observability were not supported.

Peer Influence had the largest beta coefficient which make this attribute the strongest predictor. It is equally interesting to see that the beta coefficient for Compatibility and Ease of use is also near high confirming the importance of those attributes. Over 79% felt they enjoyed the idea of programming contests and strongly agreed that it would enhance their problem solving skills. In terms of gender, over 60% of both gender felt that it was a status symbol to join programming contests. But there was much higher positive attitude among male students towards programming contests. This was possibly because how they perceive programming contests contribute towards finding better jobs as enumerated by the survey data.

Training and participation in competitive events that involve problem solving enhance multiple skills amongst its participants. Use of programming as a means to solve real-life like problems in contest environments helps rank worldwide talent. Recent years have seen progressive use of technology in programming contests in order to achieve automation and scalability. The participants are incentivized by prizes, fame enhancement of their programming skills.

We believe programming contests have much to offer in enhancing problem solving skills of students but it is important include programming problems and scenarios that simulate real-world issues (Raman et al. 2014a, b).

While we see the increasing need and growing adoption of programming contests, not all students are motivated to participate in challenging learning environments such as contests for similar reasons. However, the user interfaces and first experiences with learning programming and problem solving go a long way in sustaining student's motivation to compete. Prize award money are not necessarily the primary motivators for the teams (Verdú et al. 2012; Bloomfield and Sotomayor 2016). Instead the opportunities these contests provide to develop cross-cutting skills that are both technical and inter-personal in nature. Fisher and Cox (2006) suggest that a more inclusive and flexible structure will allow contests to be more appealing to a wider range of participants by being less biased towards specific traits. Findings of Zheng et al. (2011) suggest that a balanced view of intrinsic as well as extrinsic motivation is required to promote people participate in competitions.

In the recent years' academic institutions have started offering undergraduate degree courses like Introduction to competitive programming contests validating the need for formal training in this area. The analysis of data from the programming contests can provide inputs to CS curriculum designers especially during the time when declining student interest in STEM studies is a major challenge (Raman et al., 2014a, b). We

believe that the CS education can be positively impacted by integrating competitive programming contests as part of the curriculum. Today the domain of CS education includes the social media platforms and the way information flows through them. Hence it is imperative that student skills are enhanced with programming contests to enhance their CS skills. In the future work we plan to study influence the team cooperation and collaborative learning in programming contests (Raman et al. 2014a, b).

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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